

TTR TASK GUIDELINES

Human Study — Participant Instructions

Version 1.0 | 2026-02-25

TTR Task Guide: Collaborative Robot Building

TASK SETUP & RULES

10 Episode Session



Complete all 10 episodes in one browser session without closing the window.

15 Turn Limit



Each episode allows a maximum of 15 chat turns before automated termination.

The "DONE" Signal



Type "DONE" only when the robot's construction perfectly matches the target grid.

COMPONENTS & COLORS

Shapes



Washer



Screw



Nut



Vertical Bridge



Horizontal Bridge

Colors



Red



Blue



Green

Grid Size



Columns 1-8

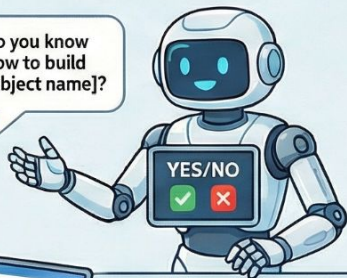
COLLABORATION WORKFLOW

VERIFY ROBOT KNOWLEDGE

Ask "do you know how to build [object name]?" before giving instructions.



do you know how to build [object name]?



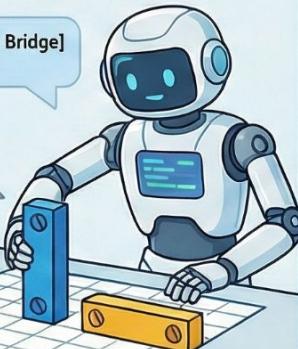
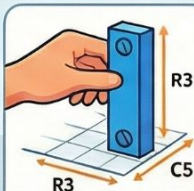
Ask "do you know how to build [object name]?" before giving instructions.

PROVIDE PRECISE INSTRUCTIONS

Specify the object name, grid coordinates (row, col), and required colors.



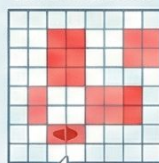
BUILD: [Blue Vertical Bridge] at [Row 3, Col 5]



MONITOR FEEDBACK TOOLS

Use the "Difference Grid" and "Reconstruction Status" flag to verify matching progress.

DIFFERENCE GRID

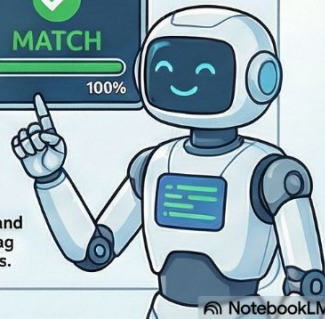


Mismatch if the robot's differs are

RECONSTRUCTION STATUS



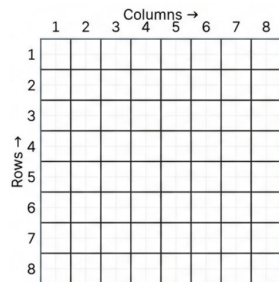
Use the "Difference Grid" and "Reconstruction Status" flag to verify matching progress.



Overview

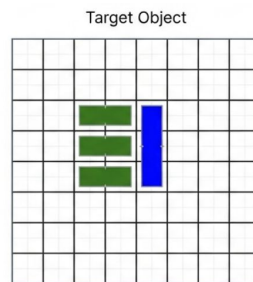
What you are doing:

- You will act as a human instructor guiding a robot to reconstruct a target object on a grid board. Your goal is to communicate clearly in English, using the chat interface.
- In each episode, you see a target object displayed on a grid and must instruct the robot to replicate that object.



User (You)

The Guide. Provide location, color, and sequence.



Partner (Robot)

The Builder. Places elements based on instructions.

Session Constraints

Session Rules:

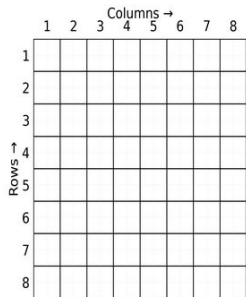
- Important: Do Not Close Your Browser until all 10 episodes are complete
- Turn Limit: Maximum 15 turns per episode.
- Auto-Abort: System loads a new episode if an irrevocable error occurs
- Completion: System stops loading new episodes when finished

Task Environment

The Board:

- The robot works on an 8×8 grid, with columns numbered 1–8 (left to right) and rows numbered 1–8 (top to bottom). The grid cells can be referred with the row, column values, for example: row 7, col 5.
- The robot can place five types of shapes, each available in four colors

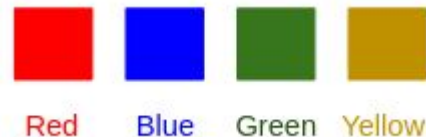
Sample Grid



Available shapes

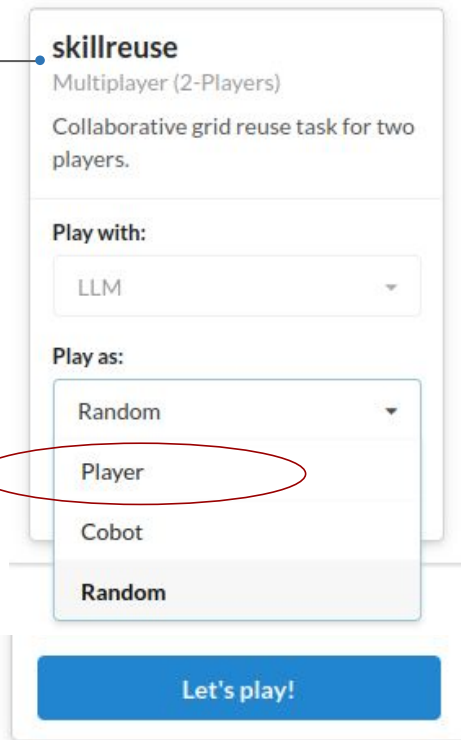


Available colors



How to Play?

1. Go to the [portal](#)
2. Select Game: SkillReuse
3. Set Role: Player
4. Click “Let’s Play!”



The image shows a configuration window for a game called "skillreuse". It is a multiplayer game for 2 players, described as a "Collaborative grid reuse task for two players." There are two dropdown menus: "Play with:" set to "LLM" and "Play as:" with a list of options: "Random", "Player", "Cobot", and "Random" (highlighted at the bottom). The "Player" option is circled in red. A blue button at the bottom says "Let's play!".

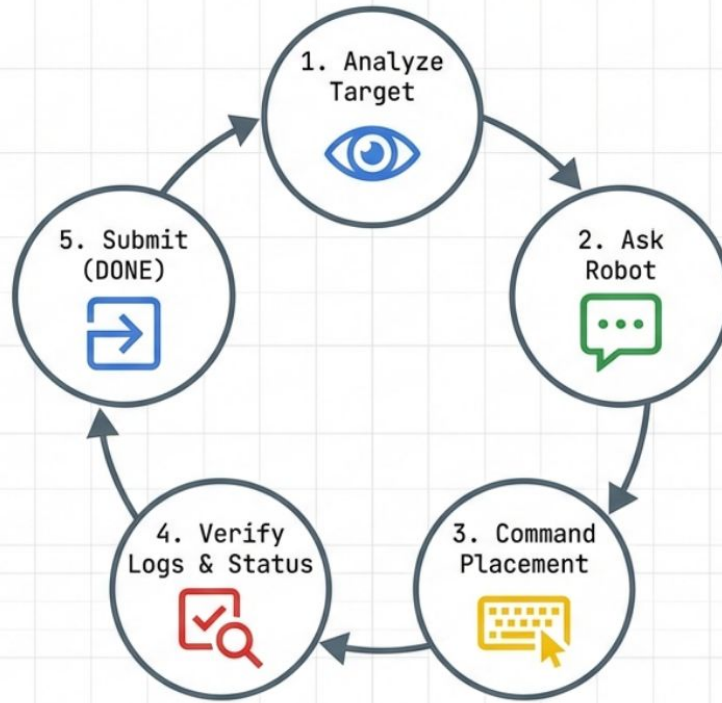
skillreuse
Multiplayer (2-Players)
Collaborative grid reuse task for two players.

Play with:
LLM

Play as:
Random
Player
Cobot
Random

Let's play!

Workflow Summary



Maintain browser session until all 10 episodes are complete.

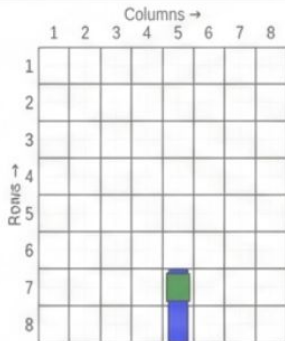
Communicating with the Robot

Grid size: 8x8
Object name: bvn
Colors: ['blue', 'green']
Location: [7, 5]
Goal:
Grid levels (bottom to top):
Level 1:
row: 7, col: 5: {'shapes': ['bvn'], 'colors': ["['blue', 'green']"]}

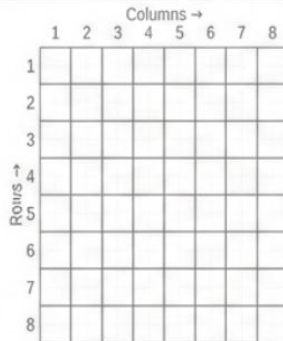
Reference Images are given below:



Goal Grid



Current Player Grid



Grid size: 8x8
Object name: bvn
Colors: ['blue', 'green']
Location: [7, 5]
Goal:
Grid levels (bottom to top):
Level 1:
row: 7, col: 5: {'shapes': ['bvn'], 'colors': ["['blue', 'green']"]}

Object Name

Required Colors

Target Location

Read the Episode Prompt

Each episode shows you the target object's name, required colors, and grid location. You need to use these while communicating with the robot.

Knowledge Check

Check the Robot's Knowledge

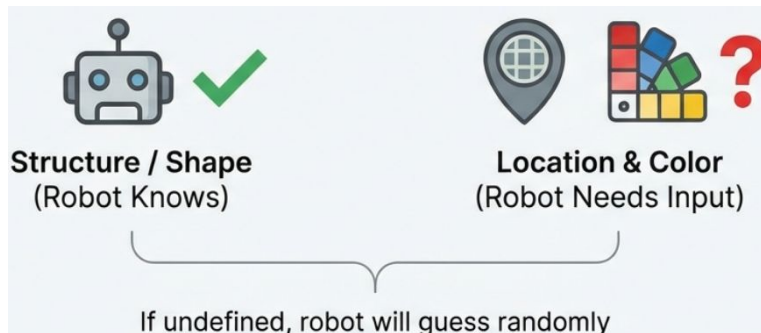
Never assume the robot knows the object. Ask first.

Before giving placement instructions, ask the robot: "do you know how to build [object name]?" The robot's response appears under Acknowledgement.

User

do you know how to build bvn?

Acknowledgement: **Yes, I know**



User

place bvn

Clarification: **Please specify the color for the 'bvn' object you want to place.**

Collaborating with the Robot

Give Placement Instructions

If the robot **knows** the object, send a single instruction specifying the object name, grid position, and colors.

User

place bvn in 7,5. Use blue, green colors

Give Placement Instructions

If the robot **doesn't know** the object, give step-by-step placement instructions

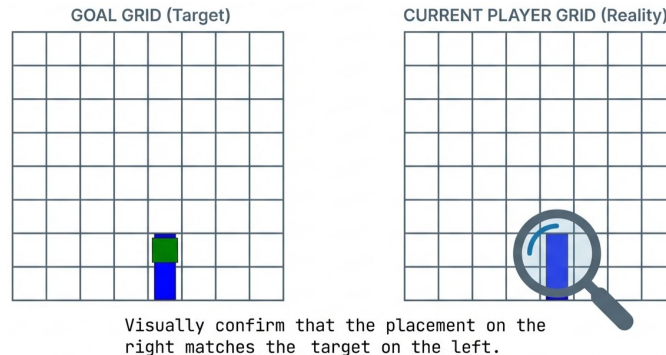
User

place a green vertical bridge
at row:7, col:5

Verification

Visual Verification

Visual verification from the player grid.



Reconstruction Status Flag

The flag shows if the robot grid matches the target grid exactly.

Reconstruction Status: **True**



Reconstruction Status: **False**

Understanding mismatches

Difference Grid

MISMATCH DETECTED

Difference grid: Differences between target grid and player's grid (bottom to top):

Level 1:

row: 7, col: 5:

- Expected: {'shapes': ['bridge-v-top'], 'colors': ['blue']}
- Current Status: {'shapes': ['bridge-v-top'], 'colors': ['green']}

row: 8, col: 5:

- Expected: {'shapes': ['bridge-v-bottom'], 'colors': ['blue']}
- Current Status: {'shapes': ['bridge-v-bottom'], 'colors': ['green']}

Level 2:

row: 7, col: 5:

- Expected: {'shapes': ['nut'], 'colors': ['green']}
- Current Status: {'shapes': ['nut'], 'colors': ['blue']}

MATCH CONFIRMED

Difference grid: Differences between target grid and player's grid (bottom to top):

Level 1:

row: 7, col: 5: Identical

row: 8, col: 5: Identical

Level 2:

row: 7, col: 5: Identical

Completion

Signalling episode completion	Type DONE (case insensitive) when the robot's grid matches with the target
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Read the Episode Prompt

- Ensure the grids match visually
- Ensure the difference grid is identical
- Ensure reconstruction status is true